PROJECT 13 • PANEL PRESENTATION DECK

Banking Loan Risk Analysis Application

Full-Stack Automated ETL, MySQL Relational Mart, Risk Bands & Streamlit Portal

Core Technologies: Python 3 • FastAPI • MySQL • Pandas • Streamlit

Key Capabilities: Automated Ingestion • 8 Data Quality Checks • MD5 Reconciliation • Explainable Low/Med/High Bands • What-If Simulator • Underwriter Review Queue

1. Problem Statement

- Financial institutions handle tens of thousands of credit lines with rapid monthly statement shifts.
- Manual, spreadsheet-based workflows are slow, prone to human error, and lack auditability.
- Underwriters struggle to identify high-exposure delinquent accounts before severe default occurs.
- Core Questions Solved: Which customer groups have higher payment risks? How do limits and bills vary across segments? Which records require additional review? What are the 6-month trends?

STRATEGIC OBJECTIVES

2. Business Need & Value Proposition

- **Automated Ingestion:** Clean and transform 30,000 credit records with record-level cryptographic reconciliation.
- **Explainable Risk Bands:** Documented thresholds for Low, Medium, High risk bands based on delay, utilization, and repayment ratios.
- **Exception Workflow:** Automatic routing of high-exposure accounts to an underwriter review queue.
- **Interactive Portal:** Streamlit dashboard providing executive KPIs, customer ledgers, and what-if simulation.

END-TO-END MICROSERVICE DESIGN

3. System Architecture & Flow

- **1. Data Ingestion:** UCI Credit Card Clients Dataset (30,000 rows) ingested via Pandas.
- **2. Data Quality & Reconciliation:** 8 automated sanity checks + MD5 checksum verification.
- **3. Relational Storage:** Normalized MySQL tables (customers, credit_profiles, bills, payments).
- **4. Risk & Rules Engine:** Multi-factor 0-100 scoring, risk band mapping, manual review triggers.
- **5. FastAPI Backend:** Paginated REST API with JWT role-based access control.
- **6. Streamlit UI:** Multi-tab executive dashboard, ledger explorer, and interactive simulator.

ARCHITECTURE SELECTION

4. Technology Stack Justification

- **FastAPI:** Asynchronous ASGI performance, automatic OpenAPI documentation, Pydantic V2 schemas.
- **MySQL (with SQLite Fallback):** Robust relational ACID storage, indexes, and seamless developer onboarding.
- **Pandas & SQLAlchemy ORM:** Efficient vector cleaning, categorical feature decoding, and relational persistence.
- **Streamlit & Plotly:** Rich interactive UI for portfolio analysis and real-time underwriter review without web frontend boilerplate.

DOCUMENTED POLICY THRESHOLDS

5. Risk Scoring & Underwriting Rules

- **High Risk Band:** Max delay ≥ 2 months, OR utilization ≥ 85% with repayment < 20%, OR default indicator = 1, OR score ≥ 65.
- **Medium Risk Band:** Max delay = 1 month, OR utilization 60%-85%, OR repayment ratio 20%-50%, OR score 35-64.
- **Low Risk Band:** Max delay ≤ 0 months, utilization < 60%, repayment ratio ≥ 50%, and score < 35.
- **Underwriter Review Triggers:** Credit limit > NT00K with delay; utilization > 90% with repayment < 15%; chronic delinquency in 2+ cycles; statement > NT0K with zero payment.

INTERACTIVE CAPABILITIES

6. Key Portal Features & Demo

- **Executive KPI Dashboard:** Real-time exposure meters, risk band donut charts, and default distributions.
- **Customer Explorer:** Search by ID, filter by risk/education/age, and inspect 6-month billing ledgers.
- **Underwriter Queue:** Review flagged accounts, inspect triggered rules, approve or reject applications.
- **What-If Simulator:** Interactive applicant form evaluating risk scores and recommendations in real time.
- **ETL Control Center:** Run sample/incremental/full ETL pipelines, inspect reconciliation and quality logs.

TESTING & DATA FINDINGS

7. Evaluation Results & Insights

- **14 / 14 Pytest Test Cases Passed (100%):** Comprehensive ETL, risk rules, DB, and API test coverage.
- **Pipeline Speed:** 200 records in 0.21s; 3,000 records in 3.19s with perfect MD5 reconciliation.
- **Education Risk:** High School segment has highest default rate (26.2%) vs Graduate School (19.2%).
- **Age Trends:** Senior age group (60+) has highest default share (28.6%), while 30-39 is lowest (21.0%).
- **Utilization Finding:** Utilization > 85% combined with delay yields a default rate exceeding 68%.

ENGINEERING LEARNINGS

8. Challenges & Roadmap

- *Challenge:* Unnormalized wide dataset format. *Solution:* Unpivoted into normalized relational child tables with compound indexing.
- *Challenge:* Aggregation latency on 30,000 rows. *Solution:* Materialized segment_kpi_summaries table dynamically refreshed during ETL.
- *Future Roadmap:* Containerize via Docker, add Kafka streaming ingestion, and train XGBoost risk model.